AROMAL A BALAN

AI/ML Engineer

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PROFESSIONAL SUMMARY

Generative AI Engineer with 3 years 9 months of experience in designing, developing, and deploying production-grade AI solutions across the full LLM lifecycle, from prototyping and orchestration to evaluation, observability, monitoring, and production deployment. Practical experience with Retrieval-Augmented Generation (RAG), agentic and multi-agent systems, AI automation, MCP-based tool calling, and context engineering. Skilled in LLM evaluation, tracing, and observability using RAGAS, DeepEval, GEPA, LangSmith, and MLflow. Experienced in translating business requirements into scalable AI solutions, collaborating with stakeholders and clients, and mentoring engineers on GenAI best practices.

TECHNICAL SKILLS

Generative AI & Agentic AI Engineering: Generative AI, LLM Application Architecture, Context Engineering, Prompt Engineering, Advanced RAG Architectures, Corrective RAG (CRAG), Self-RAG, Agentic RAG, LLM Optimization, Multi-Agent Systems, Agentic Design Patterns (ReAct, Plan-and-Execute, Reflection, Evaluator-Optimizer), Jev (System One AI Model), MCP-Based Tool Integration

AI Frameworks, Programming & Protocols: Python, Pandas, FastAPI, REST API Design, Streamlit, LangChain, LangGraph, Google Agent Development Kit (ADK), Model Context Protocol (MCP), FastMCP, Agent2Agent (A2A) Protocol, n8n, Mem0, Supermemory, Tavily, Unsloth

Machine Learning & LLM Fine-Tuning: Machine Learning, Deep Learning, Supervised Fine-Tuning (SFT), LoRA, QLoRA, Parameter-Efficient Fine-Tuning (PEFT), Adapter Tuning, Prefix Tuning, Prompt Tuning, Direct Preference Optimization (DPO), Transformer Architecture, RNN, LSTM, TensorFlow, Scikit-learn, Word2Vec

LLM Evaluation & Observability: RAGAS, DeepEval, LLM-as-a-Judge, GEPA, LangSmith, MLflow, Experiment Tracking, Prompt & Model Versioning

Databases & Vector Search: PostgreSQL, SQL, FAISS, Pinecone, Weaviate, ChromaDB, Azure AI Search

Cloud Platforms & AI Services: Microsoft Azure (Azure OpenAI, Azure AI Foundry, Azure AI Search, Azure Prompt Flow, Azure AI Speech, Azure Communication Services, Azure Logic Apps, Azure Blob Storage), Google Cloud (Cloud Run, Google Agent Platform), AWS (EC2)

DevOps & Developer Tools: Docker, Git, GitHub, Azure DevOps, Jupyter Notebook, Visual Studio Code (VS Code), Claude Code

PROFESSIONAL EXPERIENCE

Senior Systems Engineer — AI/ML & Generative AI | Infosys Limited *July 2022 – April 2026*

- Owned end-to-end development of enterprise Generative AI solutions, translating business requirements into production-ready architectures, prototypes, and scalable deployments across RAG, agentic AI, multi-agent systems, and AI automation use cases.
- Designed and deployed multiple production-grade RAG solutions using LangChain, LangGraph, and LLM orchestration frameworks, implementing document ingestion, chunking, embeddings, hybrid semantic + keyword retrieval, metadata filtering, reranking, context engineering, and citation-grounded response generation across different deployment environments.
- Developed agentic and multi-agent RAG workflows where LLM agents plan multi-step tasks, dynamically determine retrieval requirements, invoke tools through MCP-based tool calling, validate evidence against guardrails, and iteratively refine queries and responses.
- Implemented systematic LLM evaluation and observability using RAGAS, DeepEval, LLM-as-a-Judge, and GEPA, with LangSmith and MLflow for tracing, experiment tracking, performance analysis, and continuous improvement of LLM applications.

- Developed a real-time call-intelligence solution integrating Azure Communication Services, Azure Speech, and Azure OpenAI for live transcription, sentiment analysis, summarization, and automated call-closure note generation.
- Engineered an AI-powered n8n automation using Azure OpenAI and Python to translate, validate, and generate multilingual Daily PnL commentary, automating a previously manual reporting workflow and reducing operational effort.
- Hands-on experience with agentic AI and multi-agent systems using LangGraph and MCP, designing LLM-driven workflows for multi-step task planning, dynamic tool selection and invocation, contextual retrieval, intermediate-result validation, and iterative response refinement.
- Containerized and deployed Generative AI applications and LLM-based services using Docker, ensuring consistent environments, dependency isolation, and streamlined deployment across development and production.
- Collaborated directly with business stakeholders, clients, managers, and technology providers including Microsoft to gather requirements, design solutions, resolve technical issues, and drive implementations through production.
- Mentored engineers and conducted technical knowledge-transfer sessions and solution walkthroughs covering Generative AI, RAG, agentic AI, LangChain, LangGraph, MCP, evaluation, prompt engineering, automation, and project-specific implementations.

PERSONAL PROJECT

Multilingual Clickbait Detection — Agentic AI Fact-Grounding + Fine-Tuned Classifier

- Extended earlier IEEE-published clickbait-detection research into a zero-shot multilingual system, moving from a single-language LSTM classifier to a hybrid architecture designed to generalize across languages without retraining.
- Built a hybrid detection system: an Unsloth/LoRA fine-tuned multilingual classifier fused with a LangChain + LangGraph agentic AI pipeline that autonomously retrieves and verifies evidence over MCP tool calling for fact-grounding.
- Combined classifier and agent outputs via a trained decision aggregator into a calibrated, language-localized verdict; evaluated against Accuracy, F1, MAE, and Brier Score, and shipped as an interactive Streamlit app for hands-on demonstration.

Tech: Python, FastAPI, MCP, LangChain, LangGraph, Unsloth, LoRA, SFT, Streamlit, MCP, Qwen2.5-1.5B-Instruct, Tavily

AI-Powered Minutes of Meeting (MoM) — Multilingual Meeting Transcription + Automated Summarization

- Built a voice-to-Minutes-of-Meeting platform that converts audio recordings into timestamped transcripts with speaker identification and enables export as PDF and DOCX reports.
- Implemented multilingual speech transcription, speaker diarization, transcript-speaker alignment, overlap detection, and speaker participation statistics.
- Automated the generation of meeting summaries, key discussion points, decisions, and clearly assigned action items.
- Developed a transcript-based Retrieval-Augmented Generation (RAG) assistant that searches meeting content using semantic and keyword retrieval, then returns evidence-based answers with speaker and timestamp references.

Tech: Python, FastAPI, SQLite, FAISS, Faster-Whisper (Medium), Pyannote Speaker Diarization 3.1, Qwen2.5-3B-Instruct, paraphrase-multilingual-MiniLM-L12-v2, MS MARCO MiniLM-L-6-v2.

PUBLICATIONS

Clickbait Detection Using Long Short-Term Memory — IEEE | ieeexplore.ieee.org/document/9941251

EDUCATION

Master of Computer Applications — CGPA 7.95 *August 2020 – May 2022*

Amrita School of Arts and Sciences, Kochi (Amrita Vishwa Vidyapeetham)

Bachelor of Computer Applications — CGPA 8.18 *June 2017 – March 2020*

Baselios Poulose II College, Piramadom (Mahatma Gandhi University)

LANGUAGES

English (Fluent) | Malayalam (Native) | Tamil (Conversational)